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Software Requirements
Specification
for
Scalable Academic Management System (SAMS)

Version 1.0 approved
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Revision History

Name	Date	Reason For Changes	Version
Team 10	Jan 2026	Initial SRS for Scalable Academic Management System (SAMS) – Version 1.0 approved	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the requirements for the Scalable Academic Management System (SAMS), a web-based and optionally mobile-packaged platform to manage academic grade records for higher education, with focus on Kerala Technological University (KTU) official results and internal/series examinations.
The SRS covers the complete system as implemented for the CSD 334 Mini Project, including teacher grade upload workflows (PDF and Excel), unified marks storage, student dashboard for consolidated viewing, and foundational admin monitoring features.

1.2 Document Conventions

- The term “system” refers to the Scalable Academic Management System (SAMS).
- The words “shall” and “must” indicate mandatory requirements; “should” indicates desirable but not strictly mandatory behavior.
- Functional requirements are referenced as FR-XX (e.g., FR-A1), and nonfunctional requirements as NFR-XX.
- Exam types are capitalized as: “KTU Official”, “Series 1”, “Series 2”, “Internal”.
- All data fields and collection names follow lowerCamelCase or PascalCase as used in implementation design.

1.3 Intended Audience and Reading Suggestions

The intended audience includes:

- Student development team (design, implementation, and testing).
- Project guide and department mini-project evaluation committee.
- Future maintainers who may extend the system with attendance and timetable modules.
  Readers interested in business goals and scope should focus on Sections 1 and 2, technical implementers on Sections 3, 4, and 5, and evaluators/testers on Sections 4, 5, and 6 together with the appendices.

1.4 Product Scope

SAMS provides a unified, scalable platform where:

- Teachers upload KTU result PDFs and series/internal exam Excel files.
- The system parses, validates, and stores these marks into a unified marks collection.
- Students view all their grades from both sources through a single dashboard, with filtering and summary statistics.
  The main goals are to decouple grade access from KTU portal availability, reduce manual grade collation effort for faculty, and lay a modular foundation for future academic modules (attendance, timetable) using the same architecture.

1.5 References

- IEEE Std 830-1998, “IEEE Recommended Practice for Software Requirements Specifications”.
- APJ Abdul Kalam Technological University (KTU) – CSD 334 Mini Project Syllabus and Guidelines.
- Internal project specification: “Scalable Academic Management System (SAMS) – Project Info”.

2. Overall Description

2.1 Product Perspective

SAMS is a new, self-contained web application designed as a three-tier architecture (client, application, data) with optional Android packaging through Capacitor.
It does not directly integrate with KTU systems; instead, teachers manually download official result PDFs from KTU and upload them to SAMS, which parses and stores the results alongside internal exam data in a unified database.

2.2 Product Functions

At a high level, SAMS will:

- Authenticate users (students, teachers, admins) and enforce role-based access control.
- Provide teacher dashboards for KTU PDF upload, series/internal Excel upload, upload history, and report downloads.
- Parse PDF and Excel files, validate extracted records, tag them with source and exam type, and store them in a unified marks collection.
- Generate Excel reports from stored records for teacher use.
- Provide student dashboards showing consolidated grades from all sources, with exam-type filtering and summary metrics such as totals and averages.
- Provide admin dashboards with consolidated statistics and audit logs for monitoring.

2.3 User Classes and Characteristics

- **Teacher**: Faculty members responsible for uploading grades. They may have varying technical proficiency and need a simple, guided interface with clear error feedback.
- **Student**: Undergraduate or postgraduate students accessing their grades. They are typically comfortable with web/mobile interfaces and expect fast, mobile-friendly access and filtering.
- **Admin**: IT or academic staff with higher privileges to monitor system health, manage users, and review audit logs. They generally have good technical familiarity.

2.4 Operating Environment

- **Server side**: Node.js (version ≥ 18) with Express, deployed on a cloud platform (such as Render).
- **Client side**: React with Vite build system, deployed on a cloud hosting environment (such as Vercel), accessible through modern web browsers (Chrome, Firefox, Edge, Safari).
- **Database**: MongoDB Atlas (cloud-hosted) using free-tier cluster for mini-project scale.
- **File storage**: Cloud object storage (such as Cloudflare R2) for PDFs, Excel uploads, and generated reports.

2.5 Design and Implementation Constraints

- Backend implementation shall use Node.js with Express and MongoDB.
- Frontend implementation shall use React with Vite.
- JWT-based authentication and bcrypt-based password hashing shall be employed.
- Only standard PDF and Excel (.xlsx) formats are supported for grade uploads.
- System shall be deployable and functional using free-tier cloud services suitable for a mini-project scale.

2.6 User Documentation

The following user documentation will be provided:

- Quick-start guide for teachers explaining PDF and Excel upload formats and procedures.
- Short student usage guide explaining login, dashboard navigation, filtering, and printing.
- Basic admin guide describing access to statistics and audit logs.

2.7 Assumptions and Dependencies

Assumptions:

- KTU result PDFs follow a consistent tabular structure that is reliably parseable using the chosen PDF parsing library.
- Teachers can correctly prepare Excel files with the prescribed columns (RegNo, Subject, Marks, Total, Grade).
- Students and teachers have stable internet connectivity and access to modern browsers.
- Unique register numbers are available and maintained by the institution.

Dependencies:

- Availability and correct configuration of MongoDB Atlas cluster.
- Availability of object storage for file persistence.
- Availability of PDF and Excel parsing libraries (e.g., pdf-parse, xlsx) compatible with the chosen Node.js version.

3. External Interface Requirements

3.1 User Interfaces

- **Teacher Dashboard**:
  - Navigation elements for PDF Upload, Excel Upload, Upload History, and Reports.
  - PDF upload form with course, semester, and file fields, along with progress/status indicators.
  - Excel upload form with exam type selector and file input, plus validation feedback.
  - History table showing past uploads with type, course/exam, source, status, and record count.
- **Student Dashboard**:
  - Login page for register number/email and password.
  - Unified marks view showing exam type filter (All, KTU Official, Series 1, Series 2, Internal).
  - Table or card layout listing subject, marks, total, grade, exam type, and source.
  - Summary section with total marks, average, and grade trend indication.
  - Print-friendly layout for exporting grade view.
- **Admin Dashboard**:
  - Overview cards showing total users, total PDF uploads, total Excel uploads, and total records.
  - Charts or tables for uploads grouped by source (KTU Official vs Series/Internal).
  - Audit log view with filters on user, action type, date range, and source.

3.2 Hardware Interfaces

- Client side: Any standard desktop, laptop, tablet, or smartphone capable of running a modern web browser and connected to the internet.
- Server side: Cloud-hosted infrastructure with sufficient CPU, memory, and storage for mini-project scale (as provided by free-tier services).
  No special hardware devices or peripherals are required.

3.3 Software Interfaces

- **Database Interface**:
  - Communication with MongoDB Atlas using Mongoose ODM.
  - Collections include Users, Students, Courses, UnifiedMarks, UploadLogs, and AuditLogs.
- **File Storage Interface**:

    - Interaction with object storage via HTTPS APIs for uploading and downloading PDFs, Excel sheets, and reports.
- **Library Interfaces**:
    - PDF parsing via a Node.js library (e.g., pdf-parse) returning text or structured data to be mapped into marks records.
    - Excel parsing and generation via xlsx, mapping rows into internal grade structures and generating reports from stored records.

3.4 Communications Interfaces

- All client–server interactions shall use HTTP/HTTPS with RESTful endpoints.
- Authentication shall use JWT tokens transmitted in Authorization headers (Bearer scheme).
- File uploads shall use multipart/form-data requests.
- Error responses shall use JSON with HTTP status codes and descriptive messages.

4. System Features

4.1 Authentication and Authorization

4.1.1 Description and Priority

This feature provides secure login, session management, and role-based access control for teachers, students, and admins.
Priority: High, as all other system features depend on authenticated and authorized access.

4.1.2 Stimulus/Response Sequences

- User submits login credentials (email/register number and password).
    - System validates credentials, returns a JWT token, and presents the appropriate dashboard based on user role.
- User accesses a protected endpoint with a token.
    - System validates the token, authorizes the role, and either returns the requested resource or denies access.

4.1.3 Functional Requirements

- FR-A1: The system shall authenticate users via JWT-based login.
- FR-A2: The system shall hash all user passwords using bcrypt before storage.
- FR-A3: The system shall support at least three roles: Teacher, Student, and Admin.
- FR-A4: The system shall invalidate JWT tokens on explicit logout.
- FR-A5: The system shall assign an expiration time to JWT tokens (e.g., 24 hours).

- FR-A6: The system shall restrict access to protected API endpoints based on valid JWT and user role.

4.2 Teacher – KTU PDF Upload and Processing

4.2.1 Description and Priority

This feature allows teachers to upload official KTU result PDFs, which are parsed to extract grade data and stored in the unified marks collection, along with generation of downloadable Excel reports.
Priority: High, because KTU official grade data is a core information source for the system.

4.2.2 Stimulus/Response Sequences

- Teacher selects course and semester, chooses a KTU result PDF, and submits upload.
  - System validates the file type and size, stores the file, parses it, validates extracted records, tags them as “KTU Official”, stores them, updates upload history, and provides a downloadable Excel report.
- Teacher views upload history.
  - System displays previous PDF uploads along with status, record count, and timestamps.

4.2.3 Functional Requirements

- FR-T1: The system shall allow authenticated teachers to upload PDF files for specific course and semester.
- FR-T2: The system shall validate that uploaded PDF files are within an allowed size limit and are valid PDFs.
- FR-T3: The system shall parse KTU result PDFs to extract student identifiers and grades.
- FR-T4: The system shall validate that each parsed record contains required fields such as register number and grade.
- FR-T5: The system shall tag all records derived from PDFs with source “KTU Official”.
- FR-T6: The system shall store valid records from PDFs in the unified marks collection.
- FR-T7: The system shall generate an Excel report summarizing all records from a given PDF upload.
- FR-T8: The system shall maintain an upload history for each teacher including file name, course, semester, source, record count, and status.

4.3 Teacher – Series/Internal Excel Upload and Management

4.3.1 Description and Priority

This feature enables teachers to upload and manage series and internal exam marks using structured Excel files, which are validated and stored in the unified marks collection.
Priority: High, as internal assessments are frequently conducted and must be consolidated with official results.

4.3.2 Stimulus/Response Sequences

- Teacher selects exam type (Series 1, Series 2, Internal), chooses an Excel file with marks, and submits upload.
  - System validates the Excel structure and contents, tags records with exam type and source “Series/Internal”, stores them, and updates unified upload history.
- Teacher re-uploads corrected Excel file for the same exam type.
  - System processes the new file, updating or replacing previous records according to defined rules.

4.3.3 Functional Requirements

- FR-E1: The system shall allow teachers to upload Excel files for series/internal exams with required columns (RegNo, Subject, Marks, Total, Grade).
- FR-E2: The system shall validate structure of uploaded Excel files and reject files that do not match the required format.
- FR-E3: The system shall require teachers to specify exam type (Series 1, Series 2, Internal, etc.) for each Excel upload.
- FR-E4: The system shall tag all records derived from Excel files as source “Series/Internal”, with exam type stored as a separate attribute.
- FR-E5: The system shall store validated Excel records in the unified marks collection.
- FR-E6: The system shall display clear error messages if an Excel upload fails validation.
- FR-E7: The system shall show both PDF and Excel uploads together in a unified teacher upload history.

4.4 Student Unified Grade Dashboard

4.4.1 Description and Priority

This feature presents students with a consolidated view of all their grades from KTU official results and series/internal exams, including filtering and summary metrics.
Priority: High, as it is the main student-facing output of the system.

4.4.2 Stimulus/Response Sequences

- Student logs in and opens the dashboard.
  - System retrieves all records for the student from the unified marks collection and displays them organized by exam type.
- Student selects a filter (e.g., Series 1, KTU Official).
  - System updates the view to show only records matching the selected exam type.
- Student opens the summary section.
  - System displays aggregated values, such as total marks and averages across relevant records.

4.4.3 Functional Requirements

- FR-S1: The system shall allow students to log in using register number and password.
- FR-S2: The system shall display only the marks belonging to the logged-in student.
- FR-S3: The system shall show grades from both sources (KTU Official and Series/Internal) in a unified view.
- FR-S4: The system shall allow students to filter marks by exam type.
- FR-S5: The system shall provide a summary section including totals and averages derived from the student's marks.
- FR-S6: The system shall clearly indicate the source and exam type for each grade record.
- FR-S7: The system shall offer a print-friendly version of the grade dashboard.

4.5 Admin Monitoring and Audit

4.5.1 Description and Priority

This feature allows admin users to monitor system usage through statistics and audit logs, and to manage user accounts where required.
Priority: Medium for mini-project scope, but important as foundation for future extensions.

4.5.2 Stimulus/Response Sequences

- Admin logs in and opens the dashboard.
  - System shows counts of total users, PDF uploads, Excel uploads, and grade records.
- Admin views audit logs.
  - System lists user activities such as logins, uploads, and downloads with timestamps and details.

4.5.3 Functional Requirements

- FR-AD1: The system shall provide an admin role with access to system-level dashboards.
- FR-AD2: The system shall display statistics including counts of users, PDF uploads, Excel uploads, and stored records.
- FR-AD3: The system shall log actions such as login, logout, PDF upload, Excel upload, and report downloads, and make these logs viewable to admins.
- FR-AD4: The system shall allow admins to view and deactivate user accounts, subject to defined constraints.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- NFR-P1: The system should complete parsing of typical KTU result PDFs (50–100 students) within approximately 30–60 seconds.
- NFR-P2: The system should complete parsing of typical Excel files (50–100 records) within 10–20 seconds.
- NFR-P3: The system should load the student dashboard in under 2 seconds under normal network conditions.
- NFR-P4: The system should respond to login requests within approximately 1 second.

5.2 Safety Requirements

- The system shall avoid accidental deletion of grade records by requiring explicit confirmation for any destructive admin actions.
- The system shall ensure that failures in PDF or Excel parsing do not corrupt existing stored marks.

5.3 Security Requirements

- NFR-SE1: Passwords shall be stored only in hashed form using a secure algorithm (bcrypt).
- NFR-SE2: All access to grade data shall require authentication.
- NFR-SE3: Students shall not be able to view or infer other students' grades.
- NFR-SE4: Sensitive operations (uploads, downloads, admin actions) shall be logged with user identity and timestamp.

5.4 Software Quality Attributes

- **Reliability**: The system should handle invalid PDFs or Excel files gracefully with clear error messages.
- **Usability**: Teacher and student interfaces should require minimal training, with clear labels and feedback.

- **Maintainability**: The codebase should follow modular structure and consistent naming conventions to simplify future enhancements.
- **Scalability**: The architecture should allow horizontal scaling if migrated from free-tier environments to production-scale infrastructure.

5.5 Business Rules

- Only authorized teachers for a course shall upload or modify marks related to that course.
- Admin actions overriding or deactivating accounts shall be traceable via audit logs.
- Exam types shall follow institution-approved naming (e.g., “Series 1”, “Series 2”, “Internal”).

6. Other Requirements

- Database storage limits, free-tier quotas, and network constraints shall be considered when planning deployments beyond mini-project usage scenarios.
- Internationalization is not required for initial version; English language support is sufficient.
- Future modules (attendance, timetable) shall reuse the same authentication, authorization, and UI framework defined in this SRS.

Appendix A: Glossary

(Representative terms used in this SRS.)

- **Admin**: User with elevated privileges for monitoring and management.
- **Audit Log**: Record of significant system events such as logins and uploads.
- **Excel Upload**: Teacher-provided .xlsx file containing series/internal marks.
- **Exam Type**: Category of exam such as “KTU Official”, “Series 1”, “Series 2”, “Internal”.
- **KTU Official**: Grades originating from officially published KTU results.
- **Marks**: Numeric scores awarded to a student in a particular exam.
- **PDF Upload**: Teacher-uploaded KTU result PDF file.
- **RBAC**: Role-Based Access Control mechanism.
- **Unified Marks Collection**: Logical store where grades from all sources are consolidated.

Appendix B: Analysis Models

For the mini-project scope, analysis models may include:

- A high-level use case diagram showing interactions of Teacher, Student, and Admin with the system.
- A context diagram illustrating external entities (KTU as a PDF source, object storage, database).

- Simplified class or entity-relationship diagram describing Users, Students, Courses, UnifiedMarks, UploadLogs, and AuditLogs.

(Models can be attached in the project report or annexures as required.)

Appendix C: To Be Determined List

- Exact visual design specifications (colors, fonts) for dashboards.
- Detailed policy for handling re-uploads that partially overlap existing grade records (update vs replace rules).
- Thresholds and procedures for archival or deletion of old upload logs in a future production environment.

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1. srs_template-ieee.pdf